



Cybersecurity: Theory, Tools, and Practices

CS 382 L1
Spring Semester 2026

"Security is a process, not a product." – Bruce Schneier

Course Information

Start Date	End Date	Class Location	Meeting Time
Jan 12 2026	Jan 19 2026	E-003	Tues Thurs (08:30 AM-09:45 AM)
Feb 19 2026	Mar 20 2026	Tues Thurs (08:00 AM-09:00 AM)	Tues Thurs (08:00 AM-09:00 AM)
Mar 24 2026	Apr 30 2026	E-003	Tues Thurs (08:30 AM-09:45 AM)
Jan 23 2026	Feb 18 2026	E-003	Tues Thurs (08:30 AM-09:45 AM)
Feb 14 2026	Feb 14 2026	E-003	Sat (10:30 AM-01:00 PM)

Hardware/Software Prerequisites (if any):

Linux VMs (Ubuntu), Windows acceptable if WSL2 or a Linux VM (Debian/Ubuntu) is used.

Security Tools:

OWASP ZAP / Burp Suite / Microsoft Threat Modeling, Metasploit (in lab sandbox), Wireshark, tcpdump, Nmap, Nessus trial

The instructor will ask students to install and configure tools, so he needs administrative access and / or an Admin Level Access to the Lab Computers / Laptops.

Instructor will provide VM lab images/instructions

Content Area:

This is a 3+0 credit course that can serve as a CS elective for students doing CS major or minor.

Instructor Information

Instructor: Salman Khwaja Muhammad

Office Location: TBD

Email: salman.khwaja@sse.habib.edu.pk

Office Hours: TBD

Course Description

Everything in CS, from operating systems to databases, networks, and AI, relies on secure systems. Understanding cybersecurity helps students design software that's not just functional but safe and resilient against attacks. This course is an exploration of seven core domains of cybersecurity and provides a broad introduction and practical exposure to these areas, enabling students to design, evaluate, and audit secure systems. Students will engage with the following focus areas:

1. Cryptography
2. Information Security
3. Application Security
4. Network Security
5. Social Engineering (Human Factors)
6. AI Security
7. Security Audits & Compliance

The course balances theoretical frameworks with hands-on labs, where students will use industry-standard tools and techniques to detect, analyze, and mitigate threats. It culminates in a capstone project where students prepare and present a security design and audit report.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

Apply security principles (CIA, risk, threat modeling) to the requirements and design phases of the SDLC.

Implement secure coding practices and perform peer code reviews to prevent developmentstage vulnerabilities.

Use SAST, DAST, and manual tools to identify and report security flaws during testing.

Conduct full penetration tests and develop prioritized remediation plans based on findings.

Produce professional security reports aligned with compliance and ethical governance standards.

Course Aims

This course aims to provide a comprehensive understanding of cybersecurity by integrating theoretical frameworks with practical, hands-on experience. Students will engage in hands-on exercises that utilize industry-standard tools and techniques to detect, analyze, and mitigate security threats. The course emphasizes the development of practical skills alongside conceptual knowledge, preparing students to address real-world cybersecurity challenges. It concludes with a capstone project, in which students apply their learning to prepare and present a comprehensive security design and audit report.

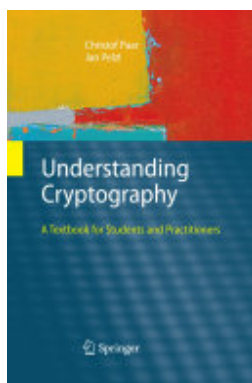
Mode of Instruction

The course is scheduled as a 75 min lectures twice a week in in-person mode. The class time will be used to discuss the topic and perform hands-on activities. All course resources (presentations, reference books, articles and all other support material) will be available on the course site on Canvas.

Engagement & Participation Rules

Students are required to attend classes regularly and actively participate in class discussions. The instructor encourages a supportive class environment that fosters Q&A, discussions, and feedback, while ensuring integrity and respect for all participants. Disrespectful behavior will not be tolerated.

Required Texts and Materials



Understanding Cryptography

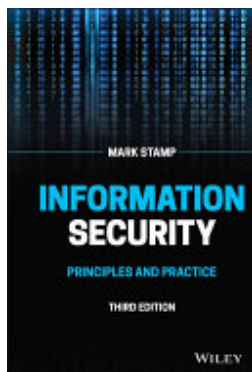
ISBN: 9783642041013

Authors: Christof Paar, Jan Pelzl

Publisher: Springer Science & Business Media

Publication Date: 2009-11-27

Optional Materials



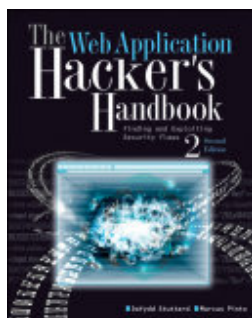
Information Security

ISBN: 9781119505907

Authors: Mark Stamp

Publisher: John Wiley & Sons

Publication Date: 2021-09-28



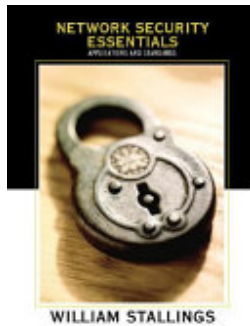
The Web Application Hacker's Handbook

ISBN: 9781118026472

Authors: Dafydd Stuttard, Marcus Pinto

Publisher: John Wiley & Sons

Publication Date: 2011-09-27



Network Security Essentials

Authors: William Stallings

Publication Date: 2000-01-01



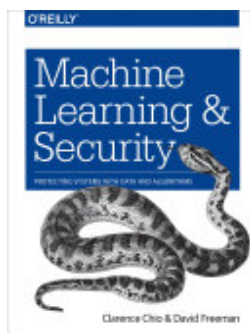
Social Engineering

ISBN: 9781119433385

Authors: Christopher Hadnagy

Publisher: John Wiley & Sons

Publication Date: 2018-07-31



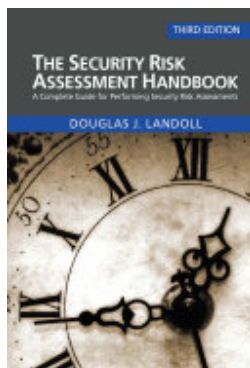
Machine Learning and Security

ISBN: 9781491979853

Authors: Clarence Chio, David Freeman

Publisher: "O'Reilly Media, Inc."

Publication Date: 2018-01-26



The Security Risk Assessment Handbook

ISBN: 9781000413205

Authors: Douglas Landoll

Publisher: CRC Press

Publication Date: 2021-09-27

Assessments

Assessment Type	Quantity	Weight (%)	Format / Remarks
Assignments	4	30	Hands-on, practical security lab reports.
Quizzes	5	10	
Midterm Exam	1	15	Covers Weeks 1–6
Project & Design Document (Report)	1	20	Individual writeup required
Final Exam	1	25	Comprehensive

Tentative list of Assignments:

- Assignment 1: Secure Application and API, due in Week 6
- Assignment 2: Network Hardening and Detection, due in week 9
- Assignment 3: AI-adversarial experiment, due in week 11
- Assignment 4: Vulnerability Scanning & Mini-audit, due in week 13
- Project & Design Document (Final Report/Audit): Week 16

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

- Students may submit one project or the report up to 48 hours late without penalty (use for emergencies only). More than one late submission will not be

accepted. Email instructor at least one hour before deadline if problem arises; communications after the deadline are not entertained.

- AI Policy: AI may be used as an assistive tool for coding, exploration, or debugging only if its use is explicitly declared and properly cited in submissions.
- AI use is strictly prohibited for the Security Reports. Any undeclared AI use for the report is treated as plagiarism. The instructor has experience recognizing AI-assisted reports.
- Failure to declare AI assistance constitutes academic dishonesty.

Week-Wise Schedule (Tentative)

Spring 2026 Weekly Schedule*

Week	Session	Readings	Assessments and Due Dates
Week - 1 January 12 – 16, 2026	Orientation & Security Thinking Mindset (Course intro; CIA triad, Risk Models)		
	Introduction to Cryptographic Concepts & Principles		
January 17, 2026	Last day to Drop Course(s)		
Week - 2 January 19 – 23, 2026	Symmetric Cryptography & Hashing (Block/stream ciphers, AES basics, use cases) Asymmetric Crypto, Signatures & PKI (RSA/ECC, digital signatures, certificate chains)		

Week - 3 January 26 – 30, 2026	Network Essentials – TCP/IP, ISO Layers, and Protocols Network Security I – Protocols & Defensive Controls (Firewalls, VPN/TLS deep dive)		Assignment 01 due
Week - 4 February 2 – 6, 2026	Network Security II – IDS/IPS & Network Capture Analysis Secure Design Principles: Threat Modeling & Secure Architecture		Quiz 01
Week - 5 February 9 – 13, 2026	Application Security I - OWASP Top 10 Deep Dive Secure Coding: Input Validation, Authentication & Session Management		
Week - 6 February 16 – 20, 2026	Application Security II – API Security (OAuth/JWT) Supply Chain Security & Dependency Management 1st Ramadant/University Switches to Ramadan Schedule: February 18, 2026		Assignment 02 due, Quiz 02
Week - 7 February 23 – 27, 2026	Static Analysis (SAST) and Peer Code Review Dynamic Analysis (DAST) & Interactive Testing (IAST/RASP)		

Week - 8 March 2 – 6, 2026	Midterm Exam (Covers Weeks 1–7) Social Engineering I – Human Factors & Psychology of Phishing		Midterm Exam
Week - 9 March 9 – 13, 2026	Social Engineering II – Pretexting & Attack Vectors AI Security I – ML Fundamentals for Security & Detection		
Week - 10 March 16 – 20, 2026	AI Security II - Adversarial ML & Fuzzing Eid ul Fitr Holidays††: March 20 – 22, 2026		Assignment 03 due Quiz 03
Week – 11 March 23 – 27, 2026	Security Audits I – Frameworks & Documentation Standards Security Audits II – Vulnerability Scanning & Reporting Pakistan Day†: March 23, 2026		Quiz 04
Week - 12 March 30 – April 3, 2026	Penetration Testing Methodology (Scoping, Recon, Exploitation) Generating Professional Security Reports (Findings, Remediation, Summary) Last Day to Withdraw from Course(s): April 3, 2026		Assignment 04 due,

Week - 13 April 6 – 10, 2026	Ethics, Legal & Responsible Disclosure in Cybersecurity Governance & Security Culture (Leadership, Awareness, Accountability)		Quiz 05
Week - 14 April 13 – 17, 2026	Capstone Planning Workshop - Integrated Architecture Design		Combine cryptography, app sec, and network defense into a unified design.
Week - 15 April 20 – 24, 2026	Integrative Security Architecture Presentations (Dry Run)		Present final system design integrating & governance, Report due
Week – 16 April 27 – May 1, 2026	Final Exam Review & Course Synthesis Last Day of Classes: April 30, 2026 Labor Day†: May 1, 2026		Review course concepts & prepare final synthesis.
May 1 – 3, 2026	Reading Days		
May 4 – 9, 2026	End Term Examinations Days§ Last Date to File Petition for Incomplete Grade: May 9, 2026		Final Exam
May 11 – 13, 2026	Faculty to discuss semester grades with students		
May 18, 2026	Final Grade Due (All Batches)		

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it

deems necessary.

† Subject to the sighting of the new moon.

‡ No class(es).

§ University's Examination Policy is available in the Academic Policies folder on the Faculty/Staff Portal.

Attendance Policy

Minimum attendance threshold: 85% of all scheduled sessions, Leaving class for > 10 minutes counts as absence.

Mandatory one-on-one meeting: each student must meet the instructor at least once before midterms (within first five weeks). Failure to meet may lead to grade penalties or academic hold.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, // bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.

- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

PLO 2: Application Development: build software systems of varying complexity in light of fundamental computer science principles and any other constraints.

PLO 4: Systems: apply the knowledge of computing systems.

PLO 7: Practical Exposure: make effective use of current tools, technologies, and good industry practices.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)					
	CLOs of the course are designed to cater following PLOs: PLO 2: Application Development PLO 4: Systems PLO 7: Practical Exposure				
	Distribution of CLO weightages for each PLO				
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 2					
PLO 4					
PLO 7					

Mapping of Assessments to CLOs

Assignments	CLO 01 (Security Thinking/Design)	CLO 02 (Secure Coding)	CLO 03 (Testing/Tools/)	CLO 04 (Evaluation/Pentest)	CLO 05 (Audit/Reporting)
Quiz 01					
Quiz 02					
Quiz 03					
Quiz 04					
Quiz 05					
Ass 01					
Ass 02					
Ass 03					

Ass 04					
Project					
Midterm Exam					
Final Exam					

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.

Safety & Ethics for Labs

Social engineering exercises must be simulated and involve only consenting participants or sandboxed simulated users. No outreach to uninformed external users, customers, or third parties is allowed unless approved in writing by the course instructor and University compliance office. All labs that could touch user data must use synthetic/test data only.

All vulnerability scanning or active testing must be performed only on lab infrastructure or systems for which explicit permission has been granted.